

Preserving Molecular Information During Pathology Report De-identification: A Synthetic Benchmark

Background: Pathology reports can support model-assisted standardization, summarization, hypothesis generation and testing, and research abstraction. De-identification must remove identifiers without erasing molecular findings.

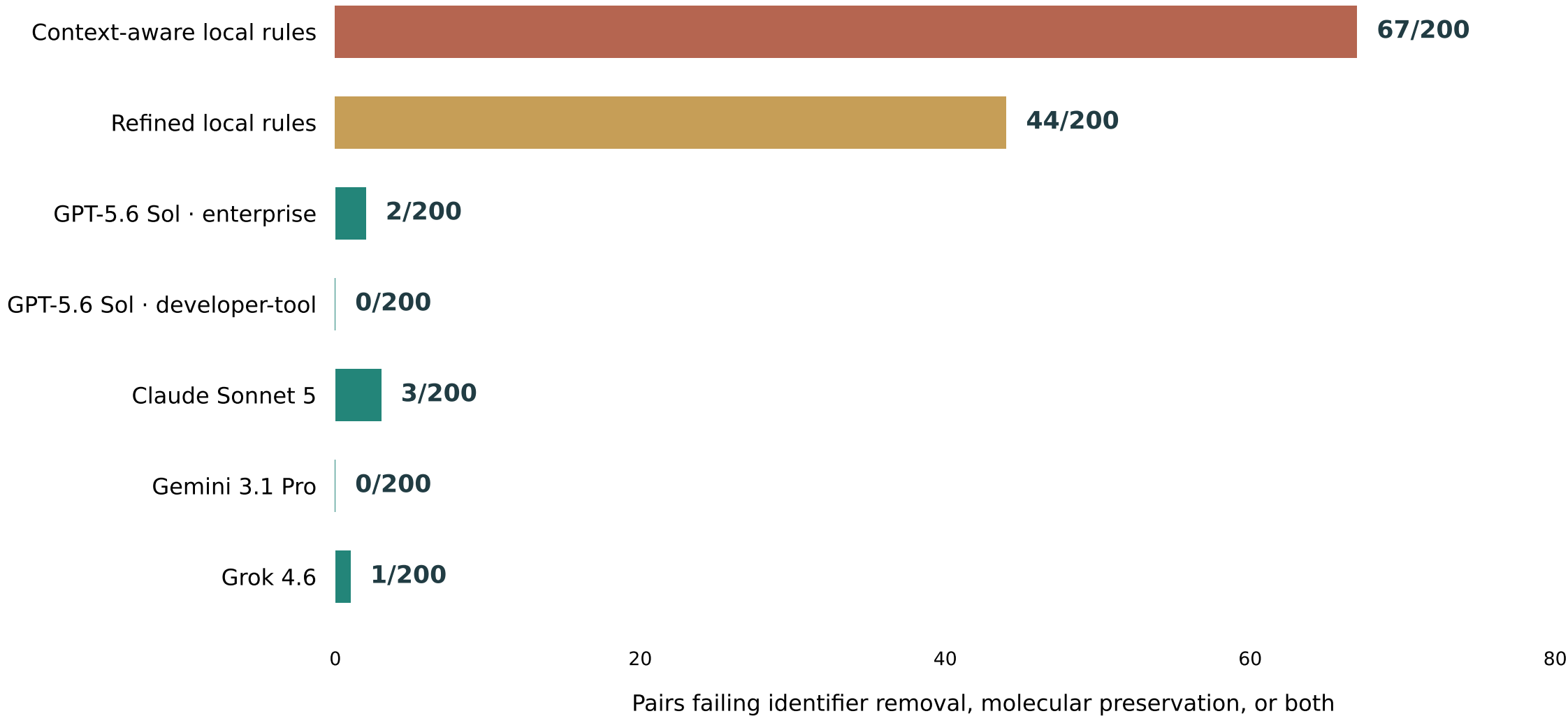
Design: We compared an institution-developed pipeline, context-aware rules, and their frozen refinement with five frontier LLM configurations (GPT-5.6 Sol through two interfaces, Claude Sonnet 5, Gemini 3.1 Pro, and Grok 4.6) on 200 unique synthetic report pairs. Joint success required preserving a molecular expression in one report and removing similarly formatted identifying digits in its paired control. A separate 200-pair experiment examined eight local failure patterns. Exploratory comparisons used paired tests and analyses grouping shared report patterns, with adjustment for multiple comparisons.

Results: Joint success was 0/200 for the institution-developed pipeline, 133/200 for context-aware rules, 156/200 for refined rules, and 197-200/200 for LLM configurations. The pipeline retained no complete molecular expressions; both local rule methods retained 159/200, versus 200/200 for every LLM configuration. Refined rules left identifying digits visible in 3/200 controls, versus 0-3/200 for LLM configurations. LLM-versus-rule differences had paired adjusted $p < 0.001$; pattern-grouped analyses were inconclusive (adjusted $p = .156$ versus earlier rules; $.375$ versus refined rules). Local failures included molecular over-redaction and identifiers missed across line breaks, separated footers, parentheses, and modified labels. In the separate experiment, refinement halved exposures from 100/200 to 50/200, with parenthetical and modified-label failures persisting.

Conclusions: Local refinement reduced identifier exposure but left substantial molecular information loss. LLM configurations achieved both objectives more consistently in this synthetic benchmark; safety and performance on independent clinical reports remain unestablished.

1. Fewer failed pairs with LLM configurations

200 unique synthetic pairs per approach | Failure of identifier removal, molecular preservation, or both



PAIR-LEVEL COMPARISON

Adjusted p < .001 versus either local rule method

ACCOUNTING FOR SHARED REPORT PATTERNS

Adjusted p = .156 / .375; evidence is inconclusive

Exploratory comparisons. Shared patterns limit the independence of the 200 pairs. Pattern-grouped values compare LLMs with earlier / refined rules. The refined comparison was added after the original analysis. The institution-developed pipeline failed all 200 pairs (Supplementary Figure S1).

A detailed synthetic biopsy and linked EGFR report, processed by unchanged local methods.

Original synthetic report

Patient name: Lena Varnell
Medical record number: 48271639
Date of birth: 06/18/1965
Ordering provider: Mira Norvale, MD
MOLECULAR PATHOLOGY REPORT
Accession: MP32-006381
Parent accession: SP32-041728
Collected: 09/10/2032 09:25
Received: 09/13/2032 11:05
Reported: 09/18/2032 14:20

SPECIMEN INFORMATION
Site: Right upper lobe lung
Diagnosis: Pulmonary adenocarcinoma
Material: Formalin-fixed paraffin-embedded core biopsy
Specimen part: A | Block: A1
Tumor cellularity: 40% | Macrodissection: Performed

MOLECULAR RESULTS
EGFR p.L858R detected; variant allele fraction 28%.
EGFR p.T790M detected; variant allele fraction 12%.

VARIANT DETAILS

Gene	Transcript	Exon	Coding change	Protein change
EGFR	NM_005228.5	21	c.2573T>G	p.L858R
EGFR	NM_005228.5	20	c.2369C>T	p.T790M

INTERPRETATION

Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS

DNA input: 42 ng | Mean target depth: 1540x
L858R locus depth: 1640x | T790M locus depth: 1240x
Target bases at >=250x coverage: 99.4%
All listed quality metrics are simulated for this illustration.

METHOD

Illustrative targeted DNA sequencing of EGFR exons 18–21.
Single-nucleotide variants and small insertions/deletions assessed.
Illustrative reporting threshold: 5% variant allele fraction.
No RNA fusion or copy-number assessment is included.

LIMITATIONS

Low tumor content, sampling, and variants below the reporting threshold may limit detection. A negative result would not exclude alterations outside the interrogated regions. No paired normal sample was analyzed. Assay specifications are illustrative only.

Electronically signed: Arlen Voss, MD
Lena Varnell | 48271639 | MP32-006381 | Page 1 of 1

Institution-developed pipeline

Patient name: **** *
***** number: *****
Date of birth: **/**/*
Ordering provider: **** *, MD
MOLECULAR PATHOLOGY REPORT
Accession: ****-*****
Parent accession: ****-*****
Collected: **/**/* 09:25
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Electronically signed: ***** *, MD
**** * | ***** | ****-***** | Page 1 of 1

Refined local rules

Patient name: *****
 ***** number: *****
 Date of birth: *****
 Ordering provider: *****

 Accession: *****
 Parent accession: *****

 Diagnosis: Pulmonary adenocarcinoma
 Material: Formalin-fixed paraffin-embedded core biopsy
 Specimen part: A | Block: A1
 Tumor cellularity: 40% | *****; Performed

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Electronically signed: ***** **, MD
 ***** | ***** | ***** | Page 1 of 1

Original synthetic report

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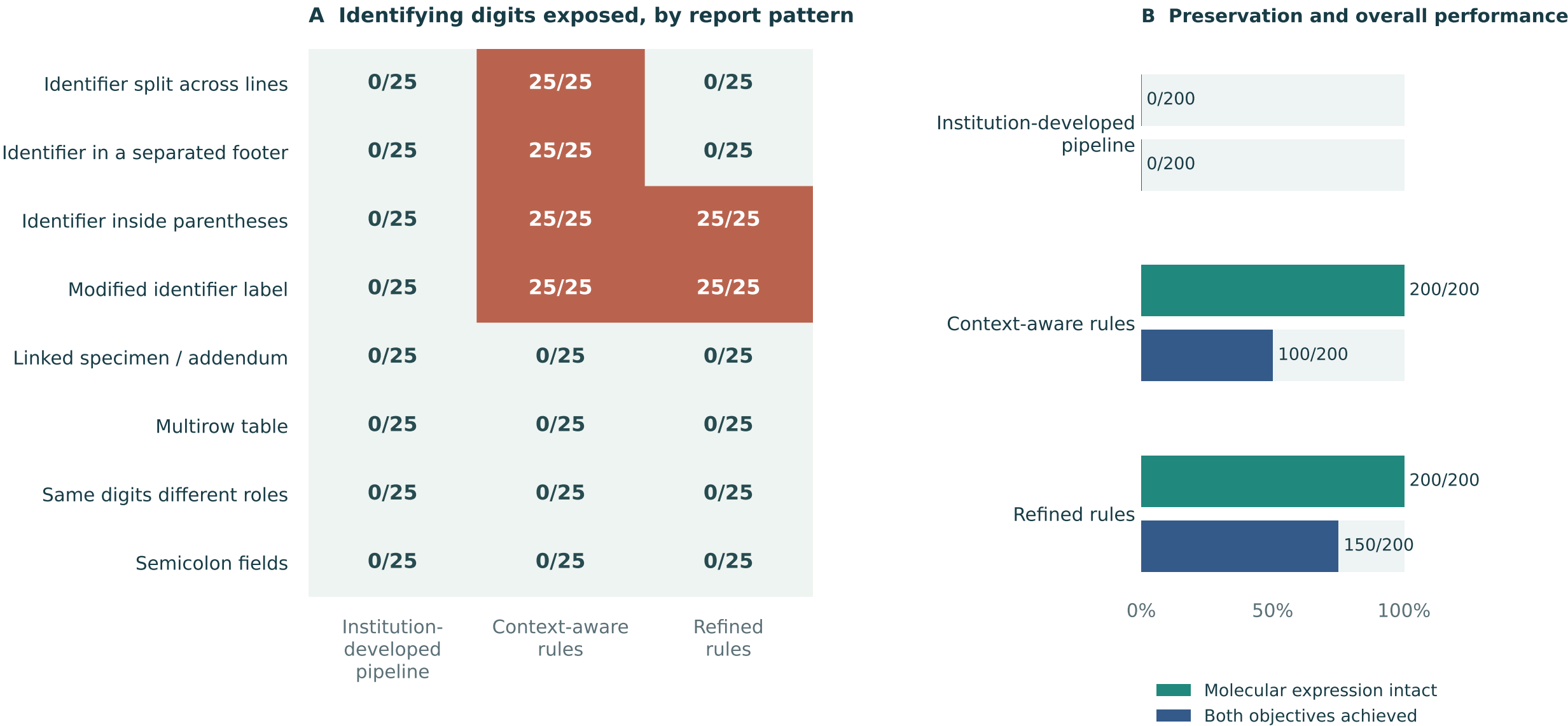
EGFR p.***** detected

EGFR p.T790M detected

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3. Local refinement: improvements and persistent failures

Separate experiment: 200 unique synthetic pairs · eight report patterns · 25 pairs per pattern



Zero tested-digit exposure alone does not imply success: the institution-developed pipeline preserved no complete molecular expressions. Refinement produced 50 additional successful pairs out of 200, across two report patterns (p = 0.50 accounting for patterns).

S1. Identifier removal and molecular preservation

Main comparison: 200 unique synthetic pairs · 400 reports per approach

